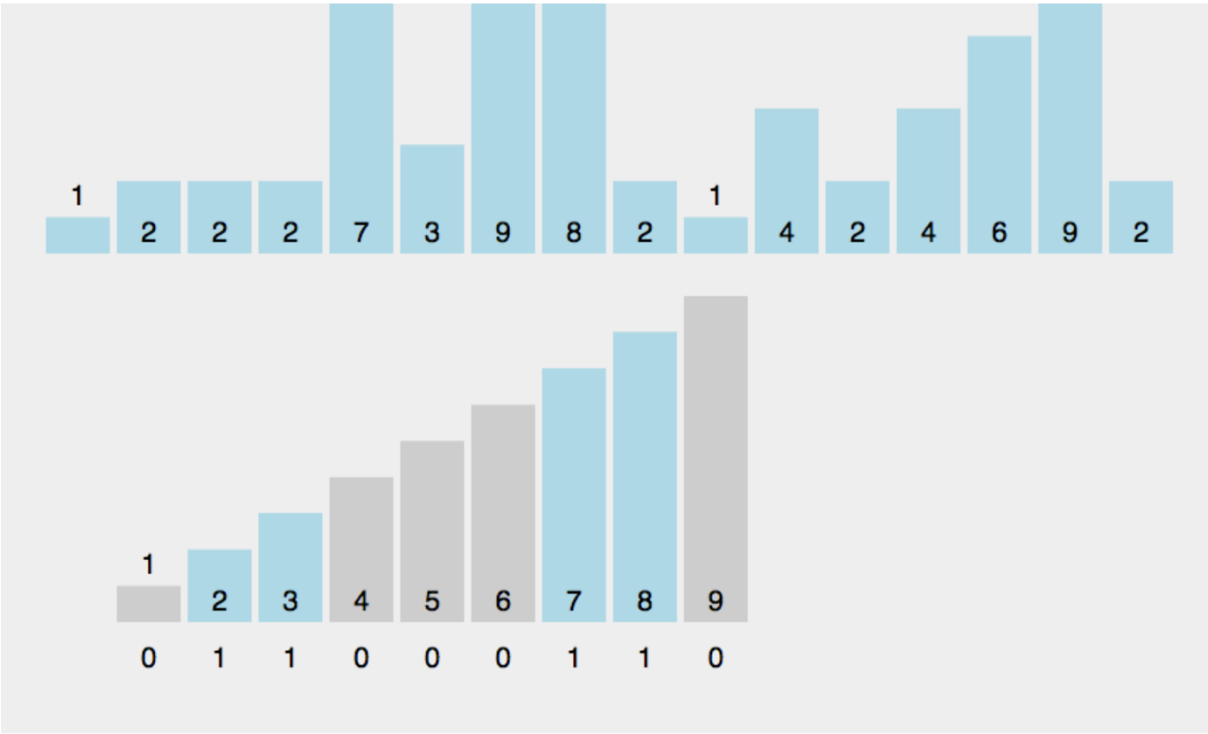
ICS4U
Unit: Algorithms

Algorithm to Code: Counting Sort

Premise is

Defined input




```
create key (counting) array

for each element in list
  increase the respective counter by 1

for each counter, starting from smallest key
  while counter is non-zero
    restore element to list
    decrease counter by 1
```

Counting Sort Algorithm & Solution
Algorithm from visualgo.net

```
def counting_sort(a_list):

    #create the key or counting array based on max value in list
    counting_list = []




    #iterate through given list and increase counters of key
    #for each element in list increase the counter by 1
    for num in a_list:



    #create a new list with values in sorted order
     based on counters
    #for each counter, starting from smallest key
    sorted_list = []
    for i in range(0, len(counting_list)):


        #while counter is non-zero
        #restore element to list and decrease counter by 1







    a_list.clear()
    a_list.extend(sorted_list)


#main
list_of_numbers = create_test_values(20, 1000)
counting_sort(list_of_numbers)
print(list_of_numbers)
```